33.

sol>

$h_n - h_{n-1} - 9h_{n-2} + 9h_{n-3} \quad (n \geq 3)$

Let $g(x) = h_0 + h_1 x + h_2 x^2 + \cdots + h_n x^n + \cdots$ be the generating function for the sequence $h_0, h_1, h_2, \cdots, h_n, \cdots$

$$
\begin{aligned}
g(x) &= h_0 + h_1 x + h_2 x^2 + h_3 x^3 + \cdots + h_n x^n + \cdots \\
-xg(x) &= \quad -h_0 x - h_1 x^2 - h_2 x^3 - \cdots - h_{n-1} x^n + \cdots \\
-9x^2 g(x) &= \quad\quad -9h_0 x^2 - 9h_1 x^3 - \cdots - 9h_{n-2} x^n + \cdots \\
+9x^3 g(x) &= \quad\quad\quad 9h_0 x^3 + \cdots + 9h_{n-3} x^n + \cdots
\end{aligned}
$$

$(1 - x - 9x^2 + 9x^3) g(x)$

$= h_0 + (h_1 - h_0)x + (h_2 - h_1 - 9h_0)x^2 + (h_3 - h_2 - 9h_1 + 9h_0)x^3$

$\cdots + (h_n - h_{n-1} - 9h_{n-2} + 9h_{n-3})x^n + \cdots$

Since $h_n - h_{n-1} - 9h_{n-2} + 9h_{n-3} = 0$

$h_0 = 0,\ h_1 = 1,\ h_2 = 2,$

We have

$(1 - x - 9x^2 + 9x^3) g(x)$

$= h_0 + (h_1 - h_0)x + (h_2 - h_1 - 9h_0)x^2$

$= x + x^2$

Thus $g(x) = \dfrac{x + x^2}{1 - x - 9x^2 + 9x^3}$

$1 - x - 9x^2 + 9x^3 = (1-x)(1-9x^2) = (1-x)(1-3x)(1+3x)$

$$\frac{x + x^2}{1 - x - 9x^2 + 9x^3} = \frac{C_1}{1-x} + \frac{C_2}{1-3x} + \frac{C_3}{1+3x}$$

$x + x^2 = C_1(1 - 9x^2) + C_2(1 + 2x - 3x^2) + C_3(1 - 4x + 3x^2)$

$= (C_1 + C_2 + C_3) + (2C_2 - 4C_3)x + (-9C_1 - 3C_2 + 3C_3)x^2$

$C_1 = -\frac{1}{4} \quad C_2 = \frac{1}{3} \quad C_3 = -\frac{1}{12}$

$$g(x) = \frac{x + x^2}{1 - x - 9x^2 + 9x^3}$$

$$= \frac{-\frac{1}{4}}{1-x} + \frac{\frac{1}{3}}{1-3x} + \frac{-\frac{1}{12}}{1+3x}$$

$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots,$

$\frac{1}{1-3x} = 1 + 3x + 3^2 x^2 + \cdots + 3^n x^n + \cdots,$

$\frac{1}{1+3x} = 1 - 3x + (-3)^2 x^2 + \cdots + (-3)^n x^n + \cdots$

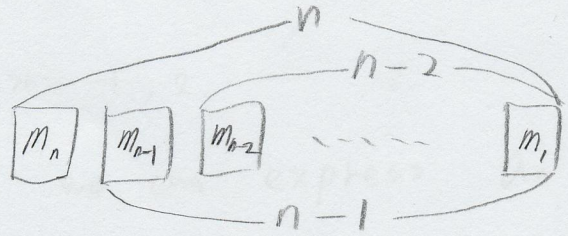
Therefore,

$$
\begin{aligned}
g(x) = &-\frac{1}{4}(1 + x + x^2 + \cdots + x^n + \cdots) \\
&+ \frac{1}{3}(1 + 3x + 3^2 x^2 + \cdots + 3^n x^n + \cdots) \\
&- \frac{1}{12}(1 - 3x + (-3)^2 x^2 + \cdots + (-3)^n x^n + \cdots)
\end{aligned}
$$

$= x + 2x^2 + \cdots + \left(-\frac{1}{4} + \frac{1}{3} \cdot 3^n - \frac{1}{12}(-3)^n\right)x^n + \cdots$

$\therefore$ We obtain $h_n = -\frac{1}{4} + \frac{1}{3} \cdot 3^n - \frac{1}{12}(-3)^n \quad (n = 0, 1, 2, \cdots)$ $\square$

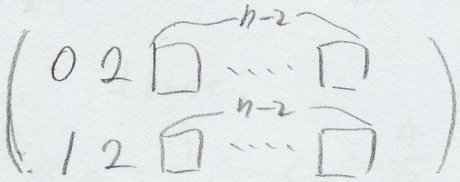
40.



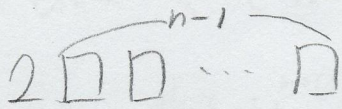
a_n contains following 3 cases, i), ii), iii)

$\left.\begin{array}{l} \text{i) } m_n = 0 \\ \text{ii) } m_n = 1 \end{array}\right\}$ m_{n-1} have to be 2. So there are

$2 \times a_{n-2}$ ways.



iii) $m_n = 2$, then there are a_{n-1} ways.



So, $a_n = \text{i)} + \text{ii)} + \text{iii)}$

$= a_{n-1} + 2 \cdot a_{n-2}$.

And, the recurrence relation is linear and homogneous. So we can let $a_n = x^n$. Then,

$x^n = x^{n-1} + 2x^{n-2}$

$x^n - x^{n-1} - 2x^{n-2} = 0$

$x^{n-2}(x^2 - x - 2) = 0$.

Since $x^{n-2} \neq 0$, $x^2 - x - 2 = 0$.

$(x+1)(x-2) = 0$.

Also, we can express a_n,

$$a_n = c_1 \cdot (-1)^n + c_2 \cdot 2^n$$

we know

$$a_0 = c_1 + c_2 = 1$$

$$a_1 = -c_1 + 2c_2 = 3$$

Solve these,

$$c_1 = -\frac{1}{3}, \quad c_2 = \frac{4}{3}$$

$$\therefore a_n = -\frac{1}{3} \cdot (-1)^n + \frac{4}{3} \cdot 2^n$$

$$= \frac{1}{3}(-1)^{n+1} + \frac{1}{3} 2^{n+2}$$

$$= \frac{1}{3}\left((-1)^{n+1} + 2^{n+2}\right).$$

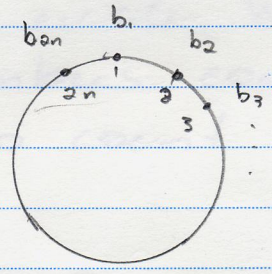
Sect 8.6

No.

1

We show a 1-to-1 correspondence between bracketings of n pairs of brackets and ways of joining the 2n points on the circle with non-intersecting lines

Given a bracketing $b_1, \ldots, b_{2n}$ where n of the b_i are '(' and n are ')', and each '(' has a paired ')', wrap the bracketing around the circle so b_i is at the ith of 2n points:



then for $i<j$ draw a line from b_i to b_j if b_i is a '(' and b_j is its paired ')'

In a bracketing we must have that if an open bracket b_j is between an open bracket b_i and its paired closed bracket b_i', then the pair of b_j must also be between b_i and b_i'. Thus none of the lines we draw will intersect

To see that a way of joining the 2n points is only constructed from one bracketing, we observe that our construction

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can be reversed:

Given a way of joining the consecutive points $1, \ldots, 2n$ on the circle with non-intersecting lines get a bracketing $b_1 \ldots b_{2n}$ by letting b_i be an open bracket if b_i is connected to b_j for $i<j$ and let it be a closed bracket otherwise.

This reverses our first construction, so we get a 1-to-1 correspondence between bracketings and ways of joining the points on the circle. We know the catalan numbers count the former, so they also count the latter.

Note: I wrote this in more detail than I would expect you to, but you should have the ideas in your proof. Several of you proved it with a correspondence using the right idea, enough to get full marks in this class. But, none of the write-ups were clear enough to serve as an example.

Several of you proved it using the hint in the book. This works also.